BMC-001	Environmental Science	L T P 0 0 0	Credits 0

Course Objectives:

CO1	To provide understanding of component of environment, their function ,quality, issues related to environment ,effect of quality degradation on human beings and their solutions.
CO2	To impart students with strong knowledge base through theory courses and sessional that makes them suitable for industries, academics, research and consultancies.
CO3	To develop students analytical, computational and research skills through assignments, weekly presentations and modeling software.
CO4	To train the students on developing practical, efficient and cost effective solutions on problems and challenges on environmental sciences and engineering.
CO5	To inculcate among students sensitivity towards social and corporate responsibilities.

MODULE 1

Introduction to Environmental Science - Definition and scope and need for public awareness Ecosystems, Concept, structure and functions, restoration of damaged ecosystems Biodiversity – Definition, description at national and global level, threats and conservation

MODULE 2

Natural Resources - Renewable and non-renewable and their equitable use for sustainability, Material cycles – carbon, nitrogen and sulphur cycle. Conventional and Non-conventional Energy Sources – fossil fuel-based, hydroelectric, wind, -nuclear and solar energy, biomass, biodiesel, hydrogen as an alternative fuel.

MODULE 3

Transportation and industrial growth Social Issues Related to Environment–Sustainable development, reset lement and rehabilitation Environmental ethics.

MODULE 4

Environmental Changes and Human Health Environmental Pollution–Definition, causes and effects, control measures for water, air, soil, noise, thermal pollution.

Textbook:

1. Environmental Studies, J Krishna wamy, RJ Ranjit Daniels, Wiley India.

Reference Books:

2. Environmental Science, Bernard J. Nebel, Richard T. Right, 9780132854467, Prentice Hall Professional 1993.
3. Environment and Ecology, RK Khandal, 978-81-265-4277-2, Wiley India.

Course OutCome:

CO1	To develop environmental scientists and engineers and sensitize them towards environmental issues.
CO2	To acquire analytical skills in assessing environmental impacts through a multidisciplinary approach.
CO3	To identify environmental problems and solutions through organized research.
CO4	To improve the communication and writing skill so as to face the competitive world
CO5	understanding of different component of environment and their function and sustainable Development